

Question 1: Full-Stack Incident Reporting & Audit Dashboard

Core Technologies

- Backend: Node.js, Express.js, PostgreSQL
- Frontend: React.js

Problem Statement

Build a secure **incident reporting system** where users can submit incidents and administrators can review, acknowledge, and resolve them through a web dashboard.

Backend Requirements

- REST API endpoints to: Create an incident, Retrieve incidents with filtering (status, category) and pagination, Update incident status (admin only)
- JWT-based authentication with roles: USER and ADMIN
- PostgreSQL database tables for: Users, Incidents, Incident status history (append-only)
- Basic input validation and error handling

Frontend Requirements

- User Interface-Incident submission form, List of user-submitted incidents with status
- Admin Dashboard-Table view of all incidents, Filters and status update controls, Manual refresh button

Implement structured audit logging using OpenTelemetry

- Every incident creation or status change must generate a trace/span
- Logs can be written to a file or database
- Explain how observability differs from simple console logging

Deliverables

- Complete backend and frontend source code
- README.md with setup and execution instructions
- Brief explanation of design choices and trade-offs
- A simple **architecture or data-flow diagram**


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Question 2: Full-Stack Secure Evidence Upload & Review Portal

Core Technologies

- Backend: Node.js, Express.js, PostgreSQL
- Frontend: React.js

Problem Statement

Build a secure web portal where users upload **evidence files** (PDF or images) linked to reports, and administrators review them.

Backend Requirements

- File upload API using multipart/form-data
- PostgreSQL tables for: Reports and File metadata
- Secure file storage (local filesystem is sufficient),
- Authorization rules: Users can access only their own files and Admins can access all files

Frontend Requirements

- **User Interface**-File upload form and List of uploaded files
- **Admin Interface**-List and preview/download evidence files

Implement envelope encryption for stored files

- Encrypt each file with a unique symmetric key
- Encrypt the file key using a master key stored in environment variables
- Explain the security benefits and trade-offs of this approach

Deliverables

- Complete backend and frontend source code
- README.md with setup and execution instructions
- Brief design explanation and trade-offs
- A simple architecture or encryption-flow diagram